Prior work and the attacker we want to study

Example: a company receives income predictions, then tries to infer sex, which the policy forbids. This is an illustrative service.

LEACE: protects released features against linear classifiers.

FARE: replaces records with cell labels. Its fairness certificate covers any classifier using those labels.

Our attacker: gets the agreed information about each person, plus labelled outputs for other people to train an inference model.

Not given: the person's original record, model weights or repeated queries.

Precedent: earlier attribute-inference attacks already learn from labelled examples.

Why study this access?

Purpose: can information provided for an allowed task also support a prohibited inference? This matters even if every defense resists.

Attack: train a classifier using other people's outputs and sensitive labels. Choose it using separate validation examples. Score it on held-out records.

Agree first: labelled outputs are a benchmark assumption. Is that access credible, and what output does the permitted task need?

Optional extension: two recipients pool outputs for the same person. We assume exact matching. No deployment with all these properties has been documented.

Appendix: different access means different attacks

Core attacker: one agreed view per person, plus labelled examples from other people.

Noise-aware attacker: also knows the noise distribution and its scale.

Clean-example attacker: also gets auxiliary features before noise was added. This is extra data, not automatically access to model weights.

Two recipients: receive both purpose outputs, matched to the same person.

Separate assumptions: repeated queries and model weights. The main classifier bank did not receive either.

Appendix: how the existing evaluation works

Provider: fits the task models and defenses.

Attacker: fits classifiers on one group, selects using a second group, and scores on a third.

Defense feedback: FARE and noise choices also used attacker-validation results.

Usefulness: some experiments measure a separate task classifier. Experiment 4 measures the model whose output is actually given to the recipient.

Reused records: all scoring records had been analysed before. These are development results, not fresh confirmation.

Appendix: what the results do and do not show

LEACE: nonlinear recovery does not contradict its linear guarantee.

FARE: its certificate concerns demographic parity, not our recovery score. Weak certificates do not prove the theorem wrong.

Stored PCRL check: 11 of 42 checks exceeded the stated threshold. This does not show that every training step violated its constraint.

Extra outputs: adding a clean task output or another recipient changes what the attacker sees.

Limits: testing a finite set of classifiers does not prove protection against every attack. No successful new algorithm has been established.